Palaeomagnetism of the Monzonite Porphyry from Milton, New South Wales

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Summary – Directions of natural remanent magnetization after *magnetic cleaning* of specimens from monzonite porphyry at Milton are given. Thermal and alternating magnetic field stability tests indicate that the resultant direction is that of the earth's magnetic field at the time of cooling. Comparison of the pole position calculated from this result with other pole positions from rocks of known age confirms the probable Permian age of the intrusion and the wide divergence of Australian from European and North American pole positions of this age.

1. Introduction

In palaeomagnetic work it is assumed that some rocks acquire an original magnetization (O) in the direction of the magnetic field at the time they were formed. Igneous rocks, cooling from a high temperature, acquire a thermoremanent magneti-

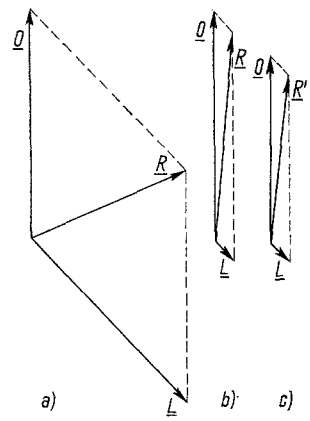


Figure 1
Showing primary (O) and secondary (L) component vectors of NRM in rocks; a) shows these components due to a partially stable rock in which both O and L are large; b) shows the components in a stable rock in which L is small; c) shows the components of rock a) after magnetic cleaning

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zation (TRM) as they pass through the Curie Point (T_c) of their magnetic minerals; the monzonite porphyry studied here has a T_c in the range 500° to 570° C. (See Figure 4). Unfortunately rocks may acquire a secondary magnetization (L) in the direction of the field at a later time. The natural remanent magnetization (NRM) is then the vector sum of the original component (O), acquired at the time of cooling, and the secondary component (L) (Figure 1), and unless L is in the same direction as O the NRM is not in the direction of the field at the time of cooling. The secondary components are commonly of viscous type, such as those examined by THELLIER [1][2]), and have a coercivity low compared with that of the original (TRM) component. THELLIER and RIMBERT [2], BRYNJOLFSSON [3], and CREER [4], have shown that such components may be preferentially removed by an alternating magnetic field, a technique that has been termed *magnetic cleaning* by AS and ZIJDERVELD [5]. After cleaning the new direction (R in Figure 1c) closely parallels the O direction and hence that of the field at the time of cooling. In this paper a further illustration of the magnetic cleaning method is given.

2. *Geology, Sampling and Statistics*

BROWN [6], has shown the monzonite porphyry at Milton, New South Wales, Australia, intrudes shattered and tilted Lower Palaeozoic remanents, and sub-horizontal Nowra Grits of the Upper Marine Series of Middle Permian age. Rocks thought to be Tertiary overlie the monzonite porphyry in places. Thus the probable limits of

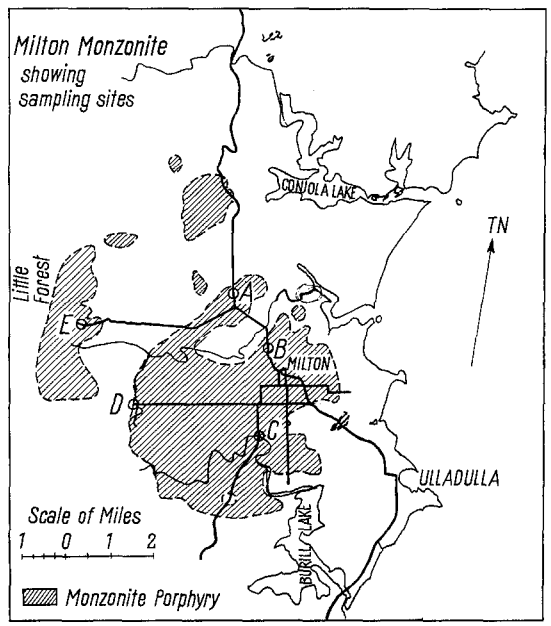


Figure 2
Sketch of the monzonite porphyry, showing sampling sites

[2]) Numbers in brackets refer to References, pages 98/99.

the age of the intrusion are Middle Permian and Tertiary. The intrusion is in the form of a laccolith with associated sills; the fact that the intruded Permian sediments have not been tilted suggests that there has been negligible movement since emplacement. The monzonite is always porphyritic, albitization is common, and much of the outcrop is weathered or soil covered. However, it was possible to obtain unweathered samples from 5 sites; a road-cutting (A), a quarry (B), two creek beds (C and D), and a road-cutting from the sill-like intrusion of Little Forest (E) (Figure 2).

The NRM directions, and the directions after *cleaning* in 225 Oe, are plotted in Figure 5. The resultant direction for each site is computed, both before and after treatment using the method of FISHER [7], and the results, together with the estimate of precision k and half-angle of the cone of confidence at $P = 0.05$, are given in Table 1. Due to the small number of samples available, specimens were used as units in the statistical analysis so that the precision given cannot be directly compared with results in which samples or sites are used as units. The two-tier analysis of WATSON and IRVING [8], gives a within site precision of 22 and a between site precision of 57 with an overall precision ($\hat{k}$) of 145, and a cone of confidence at $P = 0.05$ of 12°.

Table 1

	Initial							After magnetic cleaning in 225 Oe						
Site	S	N	D	I	R	k	a	D	I	R	k	a		
A	3	6	336	− 63	5.70	25	14	48	− 23	5.06	5.3	32		
B	3	6	34	+ 62	5.89	46	10	333	+ 77	5.69	16	17		
C	3	6	326	− 6	2.66	1.5	—	119	+ 75	5.75	20	15		
D	2	4	27	+ 46	3.83	18	23	87	+ 82	3.98	146	8		
E	1	2	—	—	—	—	—	94	+ 72	1.99	—	—	S	N
All Specimens except site A*)	3	16	11	+ 48	10.35	2.6	28	79	+ 83	17.03	17.6	9	4	18
Sites B, C, D and E as units	3	16	1	+ 39	2.36	3.1	55	85	+ 81	3.92	39	15	4	18

*) 2 specimens from site E that were not measured initially, are included in the *cleaned* results. S is the number of samples and N is the number of specimens. D is the declination clockwise from true north, and I is the inclination positive downwards. k is the precision [7] and a is the half angle of the cone of confidence at $P = 0.05$.

3. *Magnetic Cleaning*

Directions of NRM at sites A and C (see Table 1) showed a wide scatter. Four specimens from each of sites A, B and C were remeasured after treatment in each alternating magnetic field of 150, 225, 300, 450 and 600 Oe. The site directions moved away from the present earth's magnetic field direction after treatment in 225 Oe, indicating the removal of a component in this direction. In fields above 225 Oe, the directions at sites B and C did not change significantly whereas those from site A changed at each step and were random at $P = 0.05$ in fields up to 600 Oe, according

to the criterion of WATSON [9], and hence were not used. Thus the magnetic cleaning method provides a well-defined criterion for rejecting results.

Demagnetization curves for sites A, B and C are given in Figure 3. The rapid initial fall of intensity is due to the removal of a large component of low coercivity, and the initial direction at site C indicates that much of it is directed along the dipole field direction. The increase in intensity at site C after treatment in 300 Oe, and at site A after 450 Oe, is due to an increase in the resultant of two opposed vectors, since the TRM in a reversed direction is removed less rapidly than the component directed along the dipole field. This increase is comparable to that described by BOESEN, IRVING and ROBERTSON [10], for the Gingenbullen Dolerite.

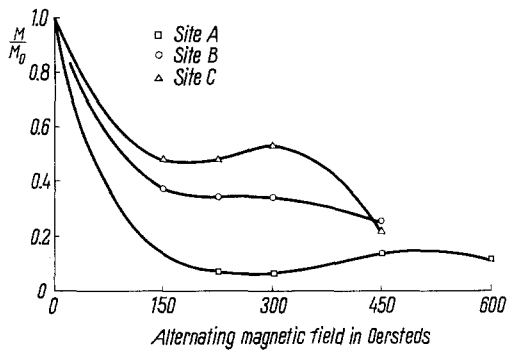


Figure 3
Alternating field demagnetization curves for sites A, B and C. The ordinate gives the ratio of intensity of magnetization remaining after treatment (M) in the peak value of the alternating field recorded along the abscissa, to the initial intensity of magnetization (M_0)

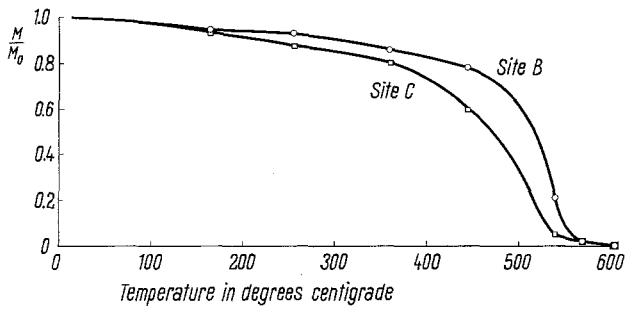


Figure 4
Demagnetization of TRM by heating for sites B and C. The ordinate has the same meaning as in Figure 3

4. *Thermoremanent Magnetization*

Four specimens from each of sites B and C were heated to 600° C and cooled in an applied field of 1.1 Oe. The TRM directions acquired were within 1° of this field direction except for one specimen, which deviated from it by 4°. This shows that the speci-

mens are magnetically isotropic and suggests that they acquired an original TRM parallel to the field in which they cooled. The specimens were then demagnetized by heating to successively higher temperatures and the normalized intensities are plotted in Figure 4. For site C the intensity falls off slowly with increase of temperature up to 400° C, for site B even more so. The curves indicate that the main Curie Points lie between 500° C and 540° C for site C, and between 530° C and 570° C for site B. In this instance the site with the higher Curie Point is the more stable but the difference is too small and the temperature spacings too wide for this to be judged significant.

5. *Discussion*

The NRM directions (Figure 5a) show a very wide scatter ($k = 3$), whereas after magnetic cleaning they are well grouped ($k = 18$, $a = 9$). Furthermore the mean direction after cleaning (85, + 81) has moved 50° of arc from the initial mean (1, + 39) in a direction away from the present field, which shows that a component in this direction has been removed. The mean direction is not significantly altered, however, after treatment in higher fields.

The distribution of sampling sites over the intrusion (Figure 2) is such that the time span covered by the specimens used should not be less than that of a secular variation cycle, [11], [12], and any secular variation component in the resultant should be small.

The pole position calculated from the mean direction, using the axial dipole field hypothesis, [13], is (32 S, 170 E) with polar errors of $dm = 24$ and $dp = 24$. It is interesting to compare this pole with those of other Australian results since it may assist in resolving the dating problems outlined in section 2.

This pole is seen in Figure 6 to be north-east of the group of Mesozoic poles of IRVING, ROBERTSON and STOTT [14], and considerably east of the Permian poles of IRVING and PARRY [15]. Despite the large error circles it seems possible that these differences are significant. There are several possible explanations:

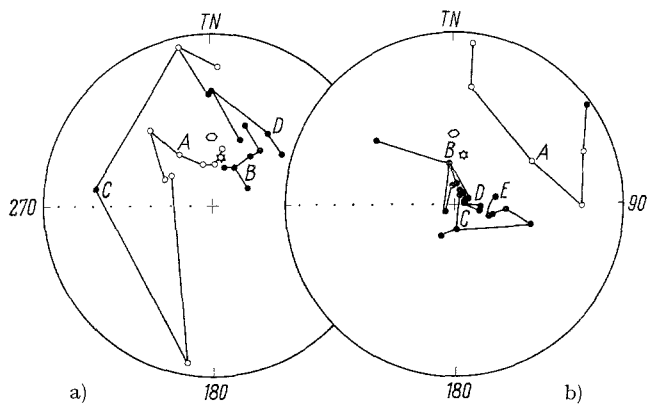


Figure 5

Directions of magnetization from specimens before a) and after b) partial demagnetization in an alternating magnetic field of peak value 225 oe. North-seeking directions are plotted as open circles on the upper hemisphere of a Schmitt equal area net, and South-seeking directions as solid circles on the lower hemisphere. The dipole (⊂) and present (☆) field directions are shown

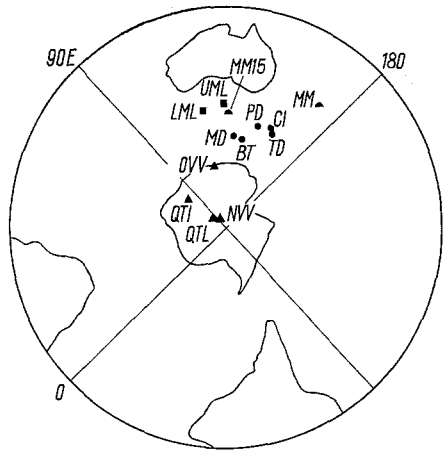


Figure 6

Comparison of the pole from the monzonite porphyry at Milton (▲) with Permian (■), Mesozoic (●) and Cenozoic (▲) Australian pole positions plotted on an equal area south polar projection. NVV = Newer Volcanics of Victoria – Plio-Pleistocene [16], QTL = Tertiary lavas from South-East Queensland – Pliocene [17], QTI = Tertiary intrusives from South-East Queensland – Mio-Pliocene [17], OVV = Older Volcanics of Victoria – Eocene [16], MD = Mount Dromedary – Lower Cretaceous – Cenomanian [18], CI = Cygnet Alkaline Complex – Lower Cretaceous – Cenomanian [19], TD = Tasmanian dolerite – Lower-Middle Jurassic [20], PD = Prospect dolerite – Lower-Middle Jurassic [10], BT = Brisbane Tuff – Middle Triassic [18], UML = Upper Marine Latites – Permian [15], LML = Lower Marine Latites – Permian [15], MM15 = Milton monzonite after a rotation of 15° to the WSW about a WNW axis

(1) The intrusion may have been tilted since emplacement by 15° to the west-south-west; such a movement would rotate the pole almost to coincide with that of the Upper Marine Latite [15], as shown in Figure 6. Field evidence does not support movement of this magnitude.

(2) Secondary components of high stability may still be present.

(3) Secular variation may not have been averaged out in the sampling.

(4) The direction may represent a field direction not previously recorded in the Permian-Cretaceous interval.

There is not yet sufficient data to discriminate between these possibilities. This result again confirms the divergence recorded previously, [14], [15], of 50° to 90° between the Australian Permian and Mesozoic poles and those from Europe and North America.

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